

Algebra (10 th std)

Choose an Appropriate Answer

1. A system of three linear equations in three variables is inconsistent if their planes

- (a) Intersect only at a point (b) Intersect in a line
(c) Coincides with each other (d) Do not intersect

Ans: (d) Do not intersect

2. The solution of the system $x + y - 3z = -6$, $-7y + 7z = 7$, $3z = 9$ is

- (a) $x = 1, y = 2, z = 3$ (b) $x = -1, y = 2, z = 3$
(c) $x = -1, y = -2, z = 3$ (d) $x = 1, y = 2, z = 3$

Ans: (d) $x = 1, y = 2, z = 3$

3. If $(x - 6)$ is the HCF of $x^2 - 2x - 24$ and $x^2 - kx - 6$ then the value of k is

- (a) 3 (b) 5 (c) 6 (d) 8

Ans: (b) 5

4. $(3y-3) / y + (7y-7) / 3y^2$ is

- (a) $\frac{9y}{7}$ (b) $\frac{9y^3}{(21y-21)}$ (c) $\frac{21y^2-42y+21}{3y^3}$ (d) $\frac{7(y^2-2y+1)}{y^2}$

Ans: (a) $9y / 7$

$$\frac{3y-3}{y} + \frac{7y-7}{3y^2} = \frac{3(y-1)}{y} + \frac{7(y-1)}{3y^2} = \frac{3(y-1)}{y} \times \frac{3y^2}{7(y-1)} = \frac{9y}{7}$$

5. $y^2 + 1 / y^2$ is not equal to

- (a) $y^4 + 1 / y^2$ (b) $(y + 1 / y)^2$ (c) $(y - 1 / y)^2$ (d) $(y + 1 / y)^2 - 2$

Ans: (b) $(y + 1 / y)^2$

$$y^2 + \frac{1}{y^2} = \left(y + \frac{1}{y}\right)^2 - 2 \times y \times \frac{1}{y} = \left(y + \frac{1}{y}\right)^2 - 2 \neq \left(y + \frac{1}{y}\right)^2$$

6. $[x / (x^2-25)] - [8 / (x^2+6x+5)]$ gives

- (a) $\frac{x^2-7x+40}{(x-5)(x+5)}$ (b) $\frac{x^2+7x+40}{(x-5)(x+5)(x+1)}$
(c) $\frac{x^2-7x+40}{(x^2-25)(x+1)}$ (d) $\frac{x^2+10}{(x^2-25)(x+1)}$

Ans:

(c) $\frac{x^2-7x+40}{(x^2-25)(x+1)}$

$$\frac{x}{x^2-25} - \frac{8}{x^2+6x+5} = \frac{x}{(x+5)(x-5)} - \frac{8}{(x+5)(x+1)} = \frac{x^2+x-8x+40}{(x^2-25)(x+1)} = \frac{x^2-7x+40}{(x^2-25)(x+1)}$$

7. The square root of $256x^8y^4z^{10} / 25x^6y^6z^6$ is equal to

(a) $\frac{16}{5} \left| \frac{x^2z^4}{y^2} \right|$ (b) $16 \left| \frac{y^2}{x^2z^4} \right|$ (c) $\frac{16}{5} \left| \frac{y}{xz^2} \right|$ (d) $\frac{16}{5} \left| \frac{xz^2}{y} \right|$

Ans:

(d) $\frac{16}{5} \left| \frac{xz^2}{y} \right|$

$$\sqrt{\frac{256x^8y^4z^{10}}{25x^6y^6z^6}} = \sqrt{\frac{256}{25} \times \frac{x^{8-6} \times z^{10-6}}{y^{6-4}}} = \sqrt{\frac{256}{25} \times \frac{x^2z^4}{y^2}} = \frac{16}{5} \left| \frac{xz^2}{y} \right|$$

8. Which of the following should be added to make $x^4 + 64$ a perfect square

(a) $4x^2$ (b) $16x^2$ (c) $8x^2$ (d) $-8x^2$

Ans: (b) $16x^2$

$$x^4 + 64 = (x^2)^2 + 8^2 + 2 \times 8x^2 = (x^2 + 8)^2$$

9. The solution of $(2x - 1)^2 = 9$ is equal to

(a) -1 (b) 2 (c) -1, 2 (d) None of these

Ans: (c) -1, 2

$$\begin{aligned} (2x - 1)^2 = 9 &\Rightarrow (2x - 1) = \sqrt{9} \Rightarrow 2x - 1 = \pm 3 \\ 2x - 1 = 3 &\Rightarrow 2x = 4 \Rightarrow x = 4/2 = 2 \\ 2x - 1 = -3 &\Rightarrow 2x = -3 + 1 \Rightarrow x = -1 \end{aligned}$$

10. If the roots of the equation $qx^2 + px + r = 0$ are the squares of the roots of the equation $qx^2 + px + r = 0$, then q, p, and r are in

(a) A.P. (b) G.P. (c) Both A.P and G.P (d) None of these

Ans: (b) G.P.

11. Graph of a linear polynomial is a

(a) Straight line (b) Circle (c) Parabola (d) Hyperbola

Ans: (a) Straight line

12. The number of points of intersection of the quadratic polynomial $x^2 + 4x + 4$ with the X axis is

(a) 0 (b) 1 (c) 0 or 1 (d) 2

Ans: (b) 1

$$(x + 2)^2 = 0 \Rightarrow (x + 2)(x + 2) = 0 \Rightarrow x = -2 \text{ or } x = -2$$

13.

For the given matrix $A = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \\ 9 & 11 & 13 & 15 \end{bmatrix}$, the order of the matrix A^T is ...

- (a) 2×3 (b) 3×2 (c) 3×4 (d) 4×3

Ans: (c) 3×4

14. If A is a 2×3 matrix and B is a 3×4 matrix, how many columns does AB have
.....

- (a) 3 (b) 4 (c) 2 (d) 5

Ans: (b) 4

15. If the number of columns and rows are not equal in matrix then it is said to be a

- (a) Diagonal matrix (b) Rectangular matrix
(c) Square matrix (d) Identity matrix

Ans: (b) Rectangular matrix

16. Transpose of a column matrix is

- (a) Unit matrix (b) Diagonal matrix (c) Column matrix (d) Row matrix

Ans: (d) Row matrix

17. Find the matrix X if

$$2X + \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 9 & 5 \end{bmatrix} \dots\dots\dots$$

- (a) $\begin{bmatrix} -2 & -2 \\ 2 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}$

Ans:

(b) $\begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix}$

18. Which of the following can be calculated from the given matrices?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix},$$

- (i) A^2 (ii) B^2 (iii) AB (iv) BA

- (a) (i) and (ii) only (b) (ii) and (iii) only
(c) (ii) and (iv) only (d) All of these

Ans: (c) (ii) and (iv) only

19. The pairs of equations $x+2y-5=0$ and $-4x-8y+20=0$ have:

- (a) Unique solution (b) Exactly two solutions
(c) Infinitely many solutions (d) No solution

Ans: (c) Infinitely many solutions

20. If a pair of linear equations is consistent, then the lines are:

- (a) Parallel (b) Always coincident
(c) Always intersecting (d) Intersecting or coincident

Ans: (d) Intersecting or coincident

21. The pairs of equations $9x + 3y + 12 = 0$ and $18x + 6y + 26 = 0$ have

- (a) Unique solution (b) Exactly two solutions
(c) Infinitely many solutions (d) No solution

Ans: (d) No solution

22. If the lines $3x+2ky-2=0$ and $2x+5y+1=0$ are parallel, then what is the value of k ?

- (a) $4/15$ (b) $15/4$ (c) $4/5$ (d) $5/4$

Ans: (b) $15/4$

$$\frac{3}{2} = \frac{2k}{5} \Rightarrow k = \frac{15}{4}$$

23. If one equation of a pair of dependent linear equations is $-3x+5y-2=0$. The second equation will be:

- (a) $-6x+10y-4=0$ (b) $6x-10y-4=0$ (c) $6x+10y-4=0$ (d) $-6x+10y+4=0$

Ans: (a) $-6x+10y-4=0$

24. The solution of the equations $x - y = 2$ and $x + y = 4$ is:

- (a) 3 and 1 (b) 4 and 3 (c) 5 and 1 (d) -1 and -3

Ans: (a) 3 and 1

25. A fraction becomes $1/3$ when 1 is subtracted from the numerator and it becomes $1/4$ when 8 is added to its denominator. The fraction obtained is:

- (a) $3/12$ (b) $4/12$ (c) $5/12$ (d) $7/12$

Ans: (c) $5/12$

26. The solution of $4/x + 3y = 14$ and $3/x - 4y = 23$ is:

- (a) $1/5$ and -2 (b) $1/3$ and $1/2$ (c) 3 and $1/2$ (d) 2 and $1/3$

Ans: (a) $1/5$ and -2

27. Rita can row downstream 20 km in 2 hours, and upstream 4 km in 2 hours. Her speed of rowing in still water and the speed of the current is:

- (a) 6 km / hr and 3 km / hr (b) 7 km / hr and 4 km / hr
(c) 6 km / hr and 4 km / hr (d) 10 km/hr and 6 km/hr

Ans: (c) 6 km / hr and 4 km / hr

28. The pair of equations $x = a$ and $y = b$ graphically represents lines which are
(a) Parallel (b) Intersecting at (b, a)
(c) Coincident (d) Intersecting at (a, b)

Ans: (d) intersecting at (a, b)

29. The pair of equations $5x - 15y = 8$ and $3x - 9y = 24/5$ has
(a) One solution (b) Two solutions
(c) Infinitely many solutions (d) No solution

Ans: (c) infinitely many solutions

30. The pair of equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ have
(a) a unique solution (b) Exactly two solutions
(c) Infinitely many solutions (d) No solution

Ans: (d) No solution

31. The value of c for which the pair of equations $cx - y = 2$ and $6x - 2y = 3$ will have infinitely many solutions is

(a) 3 (b) -3 (c) -12 (d) No value

Ans: (d) No value

32. If the lines representing the pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are coincident, then

(a) $a_1 / a_2 = b_1 / b_2$ (b) $a_1 / a_2 = b_1 / b_2 = c_1 / c_2$
(c) $a_1 / a_2 \neq b_1 / b_2$ (d) $a_1 / a_2 = b_1 / b_2 \neq c_1 / c_2$

Ans: (b) $a_1 / a_2 = b_1 / b_2 = c_1 / c_2$

33. A pair of linear equations which has a unique solution $x = 2, y = -3$ is

(a) $x + y = -1; 2x - 3y = -5$ (b) $2x + 5y = -11; 4x + 10y = -22$
(c) $2x - y = 1; 3x + 2y = 0$ (d) $x - 4y - 14 = 0; 5x - y - 13 = 0$

Ans: (b) $2x + 5y = -11; 4x + 10y = -22$

34. The father's age is six times his son's age. Four years hence, the age of the father will be four times his son's age. The present ages, in years, of the son and the father are, respectively

(a) 4 and 24 (b) 5 and 30 (c) 6 and 36 (d) 3 and 24

Ans: (c) 6 and 36

35. If the pair of linear equations has a unique solution, then the lines representing these equations will

(a) Coincide (b) Intersect at one point
(c) Parallel to each other (d) Parallel to x-axis

Ans: (b) Intersect at one point

36. Which of the following method(s) is/are used to find the solution of a pair of linear equations algebraically?

(a) Substitution Method (b) Elimination Method
(c) Cross-multiplication Method (d) All the above

Ans: (d) All the above

37. The graphical representation of a pair of equations $4x + 3y - 1 = 5$ and $12x + 9y = 15$ will be

- (a) Parallel lines (b) Coincident lines
(c) Intersecting lines (d) Perpendicular lines

Ans: (a) Parallel lines

38. Equation of $(x+1)^2 - x^2 = 0$ has number of real roots equal to:

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (a) 1

39. The roots of $100x^2 - 20x + 1 = 0$ is:

- (a) $1/20$ and $1/20$ (b) $1/10$ and $1/20$
(c) $1/10$ and $1/10$ (d) None of the above

Ans: (c) $1/10$ and $1/10$

40. The sum of two numbers is 27 and product is 182. The numbers are:

- (a) 12 and 13 (b) 13 and 14 (c) 12 and 15 (d) 13 and 24

Ans: (b) 13 and 14

41. If $\frac{1}{2}$ is a root of the quadratic equation $x^2 - mx - \frac{5}{4} = 0$, then value of m is:

- (a) 2 (b) -2 (c) -3 (d) 3

Ans: (b) -2

42. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, the other two sides of the triangle are equal to:

- (a) Base=10cm and Altitude=5cm (b) Base=12cm and Altitude=5cm
(c) Base=14cm and Altitude=10cm (d) Base=12cm and Altitude=10cm

Ans: (b) Base=12cm and Altitude=5cm

43. The roots of quadratic equation $2x^2 + x + 4 = 0$ are:

- (a) Positive and negative (b) Both Positive
(c) Both Negative (d) No real roots

Ans: (d) No real roots

44. The square root of negative number is imaginary, therefore, there is no real root for the given equation.

- (a) 4 (b) 3 (c) 3.5 (d) -3

Ans: (b) 3

45. The sum of the reciprocals of Rahman's ages 3 years ago and 5 years from now is $\frac{1}{3}$. The present age of Rahman is:

- (a) 7 (b) 10 (c) 5 (d) 6

Ans: (a) 7

46. A train travels 360 km at a uniform speed. If the speed had been 5 km / hr more, it would have taken 1 hour less for the same journey. Find the speed of the train.

- (a) 30 km / hr (b) 40 km / hr (c) 50 km / hr (d) 60 km / hr

Ans: (b) 40 km / hr

47. If one root of equation $4x^2 - 2x + k - 4 = 0$ is reciprocal of the other. The value of k is:
 (a) -8 (b) 8 (c) -4 (d) 4

Ans: (b) 8

$$\alpha \times 1/\alpha = (k-4) / 4 \Rightarrow k - 4 = 4 \Rightarrow k = 8$$

48. Which one of the following is not a quadratic equation?

- (a) $(x + 2)^2 = 2(x + 3)$ (b) $x^2 + 3x = (-1)(1 - 3x)^2$
 (c) $(x + 2)(x - 1) = x^2 - 2x - 3$ (d) $x^3 - x^2 + 2x + 1 = (x + 1)^3$

Ans: (c) $(x + 2)(x - 1) = x^2 - 2x - 3$

49. Which of the following equations has 2 as a root?

- (a) $x^2 - 4x + 5 = 0$ (b) $x^2 + 3x - 12 = 0$
 (c) $2x^2 - 7x + 6 = 0$ (d) $3x^2 - 6x - 2 = 0$

Ans: (c) $2x^2 - 7x + 6 = 0$

50. A quadratic equation $ax^2 + bx + c = 0$ has no real roots, if

- (a) $b^2 - 4ac > 0$ (b) $b^2 - 4ac = 0$ (c) $b^2 - 4ac < 0$ (d) $b^2 - ac < 0$

Ans: (c) $b^2 - 4ac < 0$

51. The product of two consecutive positive integers is 360. To find the integers, this can be represented in the form of quadratic equation as

- (a) $x^2 + x + 360 = 0$ (b) $x^2 + x - 360 = 0$
 (c) $2x^2 + x - 360 = 0$ (d) $x^2 - 2x - 360 = 0$

Ans: (b) $x^2 + x - 360 = 0$

52. The quadratic equation $2x^2 - \sqrt{5}x + 1 = 0$ has

- (a) Two distinct real roots (b) Two equal real roots
 (c) No real roots (d) More than 2 real roots

Ans: (c) No real roots

53. The equation $(x + 1)^2 - 2(x + 1) = 0$ has

- (a) Two real roots (b) No real roots
 (c) One real root (d) Two equal roots

Ans: (a) Two real roots

$$(x + 1)^2 - 2(x + 1) = 0 \Rightarrow x^2 + 1 + 2x - 2x - 2 = 0$$

$$x^2 - 1 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

54. The quadratic formula to find the roots of a quadratic equation $ax^2 + bx + c = 0$ is given by

- (a) $[-b \pm \sqrt{(b^2 - ac)}] / 2a$ (b) $[-b \pm \sqrt{(b^2 - 2ac)}] / a$
 (c) $[-b \pm \sqrt{(b^2 - 4ac)}] / 4a$ (d) $[-b \pm \sqrt{(b^2 - 4ac)}] / 2a$

Ans: (d) $[-b \pm \sqrt{(b^2 - 4ac)}] / 2a$

55. The quadratic equation $x^2 + 7x - 60$ has

- (a) Two equal roots (b) Two real and unequal roots
 (c) No real roots (d) Two equal complex roots

Ans: (b) Two real and unequal roots

56. The maximum number of roots for a quadratic equation is equal to

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (b) 2

57. If p is a prime number, then the number of factors of p is:

- (a) 1 (b) 2 (c) 3 (d) p

Ans: (b) 2

58. If one zero of the quadratic polynomial $x^2 - 4x - 5$ is 5, the other zero is:

- (a) -1 (b) 4 (c) 5 (d) 1

Ans: - 1

59. A polynomial of degree 3 is called:

- (a) Linear (b) Quadratic (c) Cubic (d) Biquadratic

Ans: (c) Cubic

60. The pair of equations $x - y = 3$ and $3x - 3y = 9$ represents:

- (a) Intersecting lines (b) Parallel lines
(c) Coincident lines (d) None of these

Ans: (c) Coincident lines

61. The pair of equations $2x + 3y = 6$ and $4x + 6y = 12$ have:

- (a) Unique solution (b) Infinitely many solutions
(c) No solution (d) None of these

Ans: (b) Infinitely many solutions

62. If the roots of a quadratic equation are equal, then the discriminant is:

- (a) Positive (b) Zero (c) Negative (d) None of these

Ans: (b) Zero

63. If the zeroes of a quadratic polynomial $ax^2 + bx + c$ are equal, then:

- (a) c and b have opposite signs (b) c and a have opposite signs
(c) c and b have same signs (d) c and a have same signs

Ans: (c) c and b have same signs

64. The quadratic polynomial whose zeroes are -3 and -4 is:

- (a) $x^2 - x + 12$ (b) $x^2 + 7x + 12$ (c) $x^2 - 7x + 12$ (d) $2x^2 + 2x - 24$

Ans: (b) $x^2 + 7x + 12$

65. The pair of equations $x = -m$ and $y = n$ graphically represent lines which are:

- (a) Parallel (b) Intersecting at $(n, -m)$
(c) Coincident (d) Intersecting at $(-m, n)$

Ans: (d) Intersecting at $(-m, n)$

66. For what value of k , do the equations $2x - y + 3 = 0$ and $6x - ky + 9 = 0$ represent coincident lines?

- (a) 2 (b) -2 (c) 3 (d) -3

Ans: (c) 3

67.If the discriminant of a quadratic equation is zero, then the roots are:

- (a) Real and unequal (b) Complex
(c) Real and equal (d) None of these

Ans: (c) Real and equal

68.If one root of $x^2 - 7x + 10 = 0$ is 5, the other root is:

- (a) 2 (b) 3 (c) 5 (d) 10

Ans: (a) 2

69.Which of the following is a factor of x^2-9 ?

- (a) $x + 3$ (b) $x - 3$ (c) $x^2 + 9$ (d) Both (a) and (b)

Ans: (d) Both (a) and (b)

70.Given $x = 2$ is a root of $p(x) = 0$, where $p(x) = x^3 - 6x^2 + 11x - 6$, then $p(2) =$

- (a) 0 (b) 6 (c) -6 (d) 2

Ans: (a) 0

71.The quadratic equation whose roots are reciprocal of the roots of $x^2-4x+1=0$ is

- (a) $x^2 - 4x + 1 = 0$ (b) $x^2 - 1 / 4 x + 1 = 0$
(c) $x^2 - 4x + 1 = 0$ (d) $x^2 - 4x + 1 = 0$

Ans: $x^2 - 4x + 1 = 0$

72.The factorization of $x^2 + 5x + 6$ is

- (a) $(x+2)(x+3)$ (b) $(x-2)(x-3)$ (c) $(x+1)(x+6)$ (d) $(x-1)(x-6)$

Ans: (a) $(x+2)(x+3)$

73.The pair of equations $3x + 6y = 9$ and $x + 2y = 3$ have

- (a) Unique solution (b) Infinite solutions (c) No solution (d) None

Ans: (b) Infinite solutions

74.The value of k for which $x^2 + kx + 9 = 0$ has equal roots is:

- (a) 18 (b) -18 (c) 6 (d) -6

Ans: (b) -18

75.The quadratic equation having 7 and - 2 as roots is

- (a) $x^2 - 5x - 14 = 0$ (b) $x^2 - 5x + 14 = 0$
(c) $x^2 + 5x - 14 = 0$ (d) $x^2 + 5x + 14 = 0$

Ans: (a) $x^2-5x-14=0$

76.The solution of $x_1 + y_1 = 5$ and $x + y = 8$ is

- (a) (2, 6) (b) (4, 4) (c) (6, 2) (d) (1, 7)

Ans: (a) (2, 6) and (c) (6, 2)

77.The highest power of x in $4x^5 - 3x^3 + 2x - 1$ is

- (a) 1 (b) 3 (c) 4 (d) 5

Ans: (d) 5

78.If the roots of $x^2 + px + 9=0$ are real and unequal, then

- (a) $p^2 > 36$ (b) $p^2 = 36$ (c) $p^2 < 36$ (d) None

Ans: (a) $p^2 > 36$

79. Solve: $3x - y = 5$ and $6x - 2y = 10$.

- (a) Unique solution (b) Infinite solutions (c) No solution (d) None

Ans: (b) Infinite solutions

80. Which of the following expressions represents a polynomial?

- (a) $x + 5$ (b) $x^3 - 5x + 1$ (c) $x+3$ (d) $x^{1/2} - 4$

Ans: (b) $x^3 - 5x + 1$

81. The zeroes of the polynomial $x^2 - 7x + 10$ are

- (a) 5 and 2 (b) 7 and 10 (c) -5 and -2 (d) None

Ans: (a) 5 and 2

82. The sum of zeroes of $4x^2 - 12x + 9$ is

- (a) 3 (b) 4 (c) 6 (d) 9

Ans: (a) 3

83. The polynomial whose degree is zero:

- (a) Constant (b) Linear (c) Quadratic (d) Cubic

Ans: (a) Constant

84. The quadratic polynomial with roots 1 and -3 is:

- (a) $x^2 + 2x - 3$ (b) $x^2 - 2x - 3$ (c) $x^2 - 2x + 3$ (d) $x^2 + 2x + 3$

Ans: (a) $x^2 + 2x - 3$

85. Which of the following equations has no solution?

- (a) $x + y = 3$ and $x + y = 5$ (b) $2x + 3y = 6$ and $4x + 6y = 12$
(c) $3x - y = 4$ and $6x - 2y = 8$ (d) None

Ans: (a) $x + y = 3$ and $x + y = 5$

86. The roots of $x^2 + x - 6 = 0$ are

- (a) 2 and -3 (b) -2 and 3 (c) 3 and -2 (d) -3 and 2

Ans: (a) 2 and -3

87. The point where two lines intersect is called the

- (a) Solution (b) Origin (c) Slope (d) Intercept

Ans: (a) Solution

88. If the roots of the quadratic equation $x^2 - 5x + 6 = 0$ are α and β , then $\alpha + \beta =$

- (a) 6 (b) 5 (c) -5 (d) -6

Ans: (b) 5

89. If p and q are prime numbers such that $p \cdot q = 15$, then $p + q = ?$

- (a) 8 (b) 7 (c) 9 (d) 10

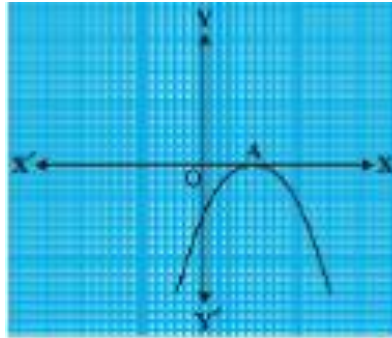
Ans: (a) 8

90. The degree of the polynomial $3x^4 - 5x^3 + 2x - 1$ is

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (d) 4

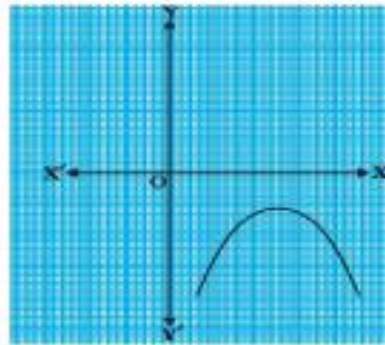
91. The number of zeroes of the polynomial from the graph is



- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (b) 1

92. The number of zeroes of the polynomial from the graph is



- (a) 0 (b) 1 (c) 2 (d) No zero

Ans: (d) No zero

93. If two lines are parallel, their system of equations has

- (a) One solution (b) No solution
(c) Infinite solutions (d) Exactly two solutions

Ans: (b) No solution

94. The graphical representation of $x + y = 4$ and $2x + 2y = 8$ are

- (a) Intersecting lines (b) Parallel lines
(c) Coincident lines (d) None of these

Ans: (c) Coincident lines

95. The solution of the system $2x + 3y = 6$; $x - y = 1$ is

- (a) (1, 2) (b) (2, 1) (c) (0, 3) (d) (3, 0)

Ans: (a) (1, 2)

96. If A is a matrix of order $m \times n$ and B is a matrix such that AB and $B^T A^T$ are both defined, then the order of matrix B is

- (a) $m \times m$ (b) $n \times n$ (c) $n \times m$ (d) $m \times n$

Ans: (c) $n \times m$

97. Choose the correct answer for the following statements.

Statement 1: Every identity matrix is a scalar matrix.

Statement 2: Every scalar matrix is a diagonal matrix.

Statement 3: Every identity matrix is a diagonal matrix.

(a) All 3 Statements are true

(b) 1 and 2 Statements are true

(c) 1 and 3 Statements are true but 2 false

(d) 2 and 3 Statements are true

Ans: (a) All 3 Statements are true

98. The matrix $A =$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

is a / an

(a) Identity matrix

(b) Symmetric matrix

(c) Skew symmetric matrix

(d) Scalar matrix

Ans: (b) Symmetric matrix

99. The matrix $A =$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is a / an

(a) Unit matrix

(b) Symmetric matrix

(c) Diagonal matrix

(d) Scalar matrix

Ans: (b) Symmetric matrix

100. If $A =$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Then $(A + I)^2 - 2A$ is equal to

(a) $2I$

(b) $3I$

(c) $-2I$

(d) I

Ans: (a) $2I$